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SHIELDING

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TRANSMISSION ENGINEERING STANDARD

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1.0 PURPOSE AND SCOPE

The purpose of this chapter is to highlight SEC practices with respect to the criteria and the method of applying shielding to minimize exposure of substation equipment and substations to direct lightning strokes.

2.0 GENERAL

2.1 Function of Shielding System

The function of a shielding system is to intercept, conduct and dissipate to ground a lightning discharge, which might otherwise strike a vulnerable part of the substation or substation equipment.

2.2 Factors Affecting Shielding Requirements

The degree of shielding required for adequate protection of the energized equipment within the substation is determined by several factors.

2.2.1 The exposure of the substation to direct lightning strokes is dependent upon the isokeraunic level (IKL) or number of storm days per year for the region. IKL applicable for SEC shall be per 01-TMSS-01.

2.2.2 The susceptibility of an object to a lightning stroke increases with the height of the object.

2.2.3 The larger the substation area, the greater the possibility the equipment to receive a direct stroke. This relationship applies not only to an individual substation but also to the total area of exposure of a power system having many substations.

2.2.4 Since no shielding system is perfect, the acceptable risk probability of a lightning stroke should also be a factor

2.3 Determination of Shielding Requirements

Statistical and probabilistic methods are used to determine the degree and type of shielding required in a given region.

2.3.1 The isokeraunic level (IKL) for a given region is based on the frequency of occurrence of thunderstorms. The average number of storm days (days on which thunder could be heard) to be expected each year in different parts of the region is used to prepare IKL charts.

2.3.2 Experimental data has been used to establish the frequency of lightning strokes to transmission lines. Analysis of this data has indicated that lightning strokes will be drawn to a transmission line, because of its height, from an effective lateral distance on each side of the line, on the average of 3.5 times the structure height.

Other statistical data has indicated that for a substation with width (*W*), length (*L*) and height (*H*) ranging from 18 to 30 meters, the total number of strokes (*P*) per year to the substation is expressed by:

$$P_{(18-30)} = K_2 \frac{(W + 2 \times 3.5H)(L + 2 \times 3.5H)}{(1609)^2} \qquad \text{(Eq.07-1)}$$

where K_2 = 9.5 for heights between 18 and 30 meters with isokeraunic level of 20 to 50.

The above equation indicates that the number of direct strokes to an unshielded substation of dimensions of 30 meters by 30 meters is approximately 0.22 per year (or one every four and a half years) in an area where the isokeraunic level is in the range of 20 to 50. Studies relating desirability of perfect shielding to the cost of shielding have indicated that a 0.1 percent exposure is a valid design criteria that would practically eliminate the possibility of direct strokes to a substation. Percent exposure for a height of 18 to 30 meters is defined as follows:

Unshielded station :	
Full exposure	one direct stroke every 8 to 10 years
Shielded station :	
Ten percent exposure	one direct stoke every 80 to 100 years
One percent exposure	one direct stroke every 800 to 1000 years
One-tenth of one percent exposure	one direct stroke every 8,000 to 10,000 years

3.0 METHODS OF SHIELDING

Effective shielding of substations may be achieved through the use of shielding masts or overhead shield wires or a combination of both. Shield wires can be used only between the terminal tower and the gantry structure. All other areas inside the substation shall be protected by a combination of gantry peaks and shielding masts.

3.1 Shielding Masts

Shielding masts can be used for all types of substations to provide protection against direct lightning strokes. They are particularly useful in large substations and those of low-profile design. They are less susceptible to mechanical failures and are preferred for substation installations.

Shielding masts shall be mounted on top of self supporting steel/wood poles or latticed-type towers and shall be electrically bonded to the substation ground grid.

3.2 Overhead Shield Wires

Overhead ground wires or shield wires are often used to provide protection against direct strokes. They can be supported by the line dead-end or terminal structures and extend over the substation up to the gantry structure.

Since these shield wires are located above the substation buses and equipment, possible breakage shall be considered in the design to avoid outage of and/or damage to these buses and equipment.

A complete overhead ground or shield wire system shall include protection for overhead lines entering and leaving the substation. In areas not employing transmission line shielding, substation shield wire systems shall be extended at least 800 meters away from the substation to limit the exposure of the phase conductors to direct strokes near the substation. For adequate protection, the transmission line overhead shield wire systems shall be directly connected to the substation shield wire system, if there is any, and electrically bonded to the substation ground grid.

4.0 ZONE OF PROTECTION

The zone of protection of a shielding system is the volume of space inside which building and equipment are considered adequately protected by the system. A shielding system allowing not more than 0.1 percent of the total predicted number of lightning strokes to terminate on protected equipment is considered adequate for most situations. For a single shielding mast, the zone of protection can be described as a cone with its apex at the highest point of the shielding mast and with a protective angle between the side of the cone and the shielding mast. For a single shield wire, the zone of protection is a wedge. When multiple shielding masts or wires are used, the zones of protection of each overlap to provide complete coverage.

Experience shows that a shielding mast or wire cannot be relied upon to provide complete protection within any particular zone, but it can be stated that the protection afforded by a shielding conductor increases as the assumed protective angle decreases.

As far as possible control building lightning protection shall be covered by lightning or lighting masts and only for buildings located away from the main electrical equipment, shielding protection in accordance with ANSI/IEEE 141 or NFPA 780 shall be provided. Lightning protection of power transformers shall be covered by lighting or lightning masts even when located adjacent to control buildings and not by air terminal located on the building. For other equipment with height considerably lesser than the height of building, lightning protection by air terminal can be considered subject to SEC approval. When lightning protection of equipment is covered with air terminal, necessary calculations based on the curves per Figure 07-1 shall be submitted.

5.0 HEIGHT AND LOCATION OF SHIELDING

The curves in Figure 07-1 were drawn using extensive laboratory test data to show the configuration of masts or overhead shield wires necessary to reduce an object's exposure to 0.1 percent. The curves are plotted to show the height (y) of the shielding masts or shield wires above the protected object as a function of the horizontal separation (x) and the height (d) of the protected object.

The dotted line curves of Figure 07-1 (a) showing the necessary configuration of a single mast protecting a single object having single prominent projection or several projections in a limited region. The full-line curves of Figure 07-1 (a) are applicable for a ring of objects distributed at a given height. The configuration of mast shall be based on most remote object. Figure 07-1 (b) and 07-1 (c) are applicable for single ground wire and two masts or two ground wires respectively. Necessary extrapolation of the curves in Figure 07-1 shall be carried out for other values of “x”, “y” and “d”.

Figures 07-2(a) through 07-2(d) illustrate the area that can be protected by two or more shielding masts. For given values of "d" and "y", values of the separation distance "s" and radii "x" can be determined from Figures 07-1(c) and 07-1(a), respectively, which give an exposure of 0.1 percent. For rectangular substations, the arrangement of two masts in Figure 07-2(a) will leave some points with a higher exposure than desired. If the distance between the masts is decreased, the resulting protected area will still be at least equal to the sum of the areas shown in Figure 07-2(a). For example, if the distance between masts in Figure 07-2(a) is halved, the resulting protected area will be larger, approximately equal to that shown in Figure 07-2(b).

On this basis, the approximate width of the overlap between masts can be obtained by taking the value of "y" from Figure 07-1(c), which corresponds to twice the actual distance between the masts. The width of overlap then equals the value of "x", obtained from Figure 07-1(a), that correspond to this "y". This gives a conservative width of the substation that can be protected by two masts.

The protected areas for three masts located at the points of an equilateral triangle or for four masts located at the points of a square are shown in Figures 07-2(c) and 07-2(d). The height of the shielding mast should be chosen so that the "b" points provide 0.1 percent exposure as obtained from Figure 07-1(c) for the mid-point between two masts. The "x" radii are obtained from the data for a single mast shown in Figure 07-1(a).

6.0 SHIELDING MATERIALS

6.1 Shielding Masts

The size of shielding masts shall be determined primarily by the mechanical strength requirements. Although the amplitude of the lightning current wave may be very high, its duration is so short that the thermal effect on a lightning protective system is usually negligible.

6.1.1 When structural steel dead-end towers are used to support shielding masts, the shielding mast shall be galvanized steel pipe with a minimum size of 38 mm diameter and 3050 mm length.

6.1.2 When wooden poles are used as support, the shielding mast shall be galvanized ground rod of 13 mm diameter and 600 mm length.

6.1.3 The shielding mast or lightning rod for the substation building shall be copper-clad ground rod of 13 mm diameter and exceeding 600 mm height.

6.1.4 When lighting metal poles are used as shielding masts, the minimum size shall be determined by the mechanical strength requirements.

6.2 Shield Wires

The shield wire material specification shall be in accordance with the requirement of 10-TMSS-03, in coordination with the material of the overhead transmission line shield wires.

6.2.1 When it is necessary to use wire shielding in a substation, the wire shall be high strength galvanized steel wire unless the available size is inadequate for current capacities or the steel wire is susceptible to corrosion in any particular application.

6.2.2 Aluminum-clad steel wire shall be used only when it is required for higher current capacity near large power sources or if it is found necessary in the coastal and contaminated areas where the wet salt spray is present in the atmosphere.

7.0 SHIELD SYSTEM GROUNDING

A shielding system cannot effectively protect substation equipment unless adequately grounded. Multiple low impedance connections from the shielding system to the substation ground grid are essential. It is beneficial to use at least two separate connections to ensure continuity and reliability. Whenever non-conducting supports are used, separate ground conductors to establish a direct connection shall be installed from the shielding system to the substation ground grid.

7.1 Shielding masts supported by steel structure shall be electrically bonded to the structure which is solidly grounded to the substation ground grid.

7.2 Overhead ground or shield wires terminated at steel structures shall be electrically bonded to the structure which is solidly grounded to the substation ground grid. The mechanical dead-end hardware shall not be used as the electrical path for conducting lightning surges or fault currents to the dead-end structure (see Figure 07-5).

7.3 When the support structure is constructed of wood or other non-conducting material, a continuous electrical path shall be made between the overhead shielding mast or wire and the station ground grid with a down conductor.

7.4 Down Conductor

7.4.1 The down conductor of a shielding mast or wire system shall be stranded bare soft drawn copper. The size of down conductor shall be 70mm² (2/0 AWG).

7.4.2 No bend of a down conductor shall form an angle of less than 90° nor shall it have a radius of bend less than 200 mm.

7.4.3 Down conductors shall be securely attached to their support structures at intervals not exceeding 2,000 mm. The fastener material shall be of such nature that there will be no serious tendency towards electrolytic corrosion.

7.4.4 Down conductors are not required for electrically continuous metal structures, but do require ground terminals. Connections to steel structures shall be made on cleaned areas of the steel framework with bolted connectors.

7.4.5 Down conductors shall be guarded/protected to prevent physical damage or displacement for a minimum distance of 2400 mm above ground level.

8.0 EXAMPLE : USE OF WORKING CURVES

The following calculation illustrates the use of empirical curves of Figure 07-1 for the substation layout given in Figure 07-3 and 07-4.

Shielding protection shall be provided for the following areas of substation as indicated in the enclosed Figure 07-3 and 07-4:

230kV Gantry Take-Off Structure Area
230kV GIS Building
230/69kV Power Transformer Area
69/13.8kV Power Transformer Area
Building Area (230kV GIS building, 69kV GIS building, Control Building and Pump House)
Other Area

230kV Gantry Take-Off Structure Area

Shielding Data:

i.	Height of Gantry	= 22.86m
ii.	Height of steel pipe considered Gantry top	= 4.05m
iii.	Height of Gantry with steel pipe	= (22.86 + 4.05) = 26.91m
	Height of objects in the area	
	230kV GIS air bushing	= 7.2m
	230kV CCVT	= 6.12m
	Hence maximum height of structure in the area to be considered for shielding protection (d)	= 7.2m
	Value of "y"	= (h-d) = (26.91-7.55) = 19.39m

Results from the curve:

From Figure 07-1 (a) and 07-1 (c), it is found the value of "x" and 's" is 23.2m and 81m respectively for the above "y" and "d" values and from the enclosed Figure 07-3 protection zone due to the same provides adequate protection.

230kV GIS Building

Shielding Data :

i.	Height of Lightning Mast	= 25m
ii.	Height of steel pipe considered on the mast top	= 2m
iii.	Total height of mast with steel pipe (h)	= (25 + 2) = 27m
iv.	Height of Gantry	= 22.86m
v.	Height of steel pipe considered Gantry top	= 4.05m
vi.	Height of Gantry with steel pipe	= (22.86 + 4.05) = 26.91m
vii.	Maximum height of GIS building (d)	= 10m

As the height of lightning mast and gantry are different, considering the minimum of the two heights for lightning protection of GIS building,"y" = (h-d)
= (26.91-10)
= 16.91m

From Figure 07-1 (a) and 07-1 (c), it is found the value of "x" and "s" is 18.3m and 76m respectively for the above "y" and "d" values. From the zone of lightning protection due to Gantry and Lightning Mast i.e total four masts, as shown in Figure 07-3, it shall be noted that lightning protection due to four masts cover lightning protection of 230kV GIS building.

230/69kV Power Transformer Area

Shielding data:

i.	Height of Lightning Mast	= 25m
ii.	Height of steel pipe considered on the mast top	= 2m
iii.	Total height of mast with steel pipe	= (25 + 2) = 27m
iv.	Height of Gantry	= 22.86m
v.	Height of steel pipe considered Gantry top	= 4.05m
vi.	Height of Gantry with steel pipe (h)	= (22.86 + 4.05) = 26.91m
vii.	Maximum height of structure in the area to be considered for shielding protection (d)	= 7.55m(230/69kV Power Transformer)
	Hence value of "y"	= (h-d) = (26.91-7.55) = 19.36m

Results from the curve:

From Figure 07-1 (a) and Figure 07-1 (c), it is found that value of "x" and "s" is more than 23.2m and 81m respectively. However to draw uniform zone of lightning protection, value of "x" and "d" are considered equivalent to 23.2m and 81m respectively. From the zone of lightning protection due to Gantry and Lightning Mast i.e total four masts, as shown in drawing Figure 07-3, it shall be noted that lightning protection due to four masts cover 230/69kV Transformer Area.

69/13.8kV Power Transformer

Shielding data:

i.	Height of Lightning Mast	= 25m
ii.	Height of steel pipe considered on the mast top	= 1m
iii.	Total height of mast with steel pipe (h)	= (25 + 1) = 26m
iv.	69/13.8kV Power Transformer maximum height to be considered for lightning protection (d)	= 4.46m
	Hence value of "y"	= (h-d) = (26-4.4) =21.54m

Results from the curve:

From Figure 07-1 (a) for the above "y" and "d", value of "x" is 27m. From the enclosed drawing Figure 07-4, it shall be noted that the protection zone due to 27 meter radius covers lightning protection for 69/13.8kV Power Transformer.

69kV GIS Building/Control Building/13.8kV Building

Lightning protection for these buildings are provided per NFPA 780 with 1000mm copper-clad air terminals installed at an interval of 7.5m and interconnected by 70 mm^2 stranded copper conductor and then the whole system is connected to main substation grounding grid, using 70mm^2 stranded copper conductor at various points with the spacing between any two grounding conductor is not exceeding 30m. Further wherever building width is more than 15.25m additional air terminal per NFPA on the middle of the roof shall be provided.

Other Area

i. Station Service Transformer

Shielding Data:

Height of Control Building	= 6.5m
Height of air terminal	= 1m
Total Height (h)	= (6.5+1)
	= 7.5m
Height of Station Service Transformer (d)	= 2.3m
Hence value of "y"	= (7.5-2.3)
	= 5.2m

From fig 07-1 (a) value of "x" is 6m. From Figure 07-4 it shall be noted that protection zone with 6m radius covers lightning protection of station service transformers.

ii. Chiller Units

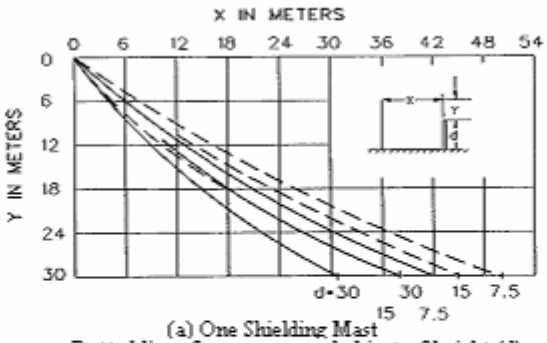
Near by lighting poles are used for lightning protection

Shielding Data:

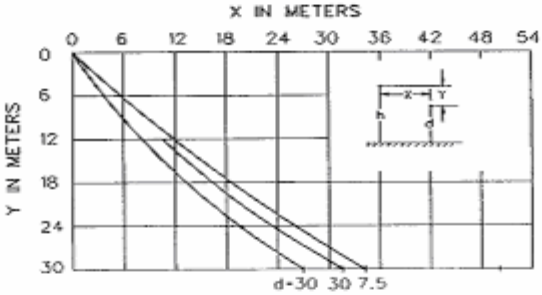
Height of lighting poles (h)	= 6.5m
Chiller Height (d)	= 2.44m

Hence value of "x" and "s" from Figure 07-1 (a) and 07-1 (c) are 4m and 32m respectively.

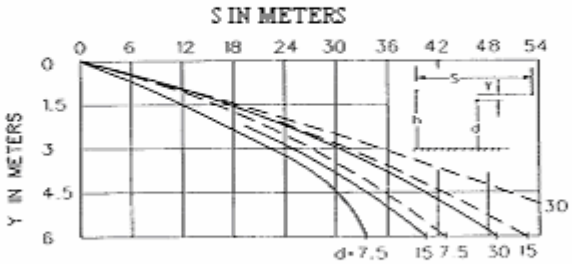
From the enclosed drawing Figure 07-4, it shall be noted zone of lightning protection due to three number lighting poles located at distance of 30.5m covers lightning protection for Chiller Units.



(a) One Shielding Mast
Dotted lines for one exposed object of height (d)
Full lines for ring of exposed objects of height (d)



(b) One horizontal ground wire



(c) Two masts or two ground wires
Dotted lines for masts. Full lines for horiznotal wires

Figure 07-1: Configuration of Shielding Object to Protected Object for 0.1 Percent Exposure

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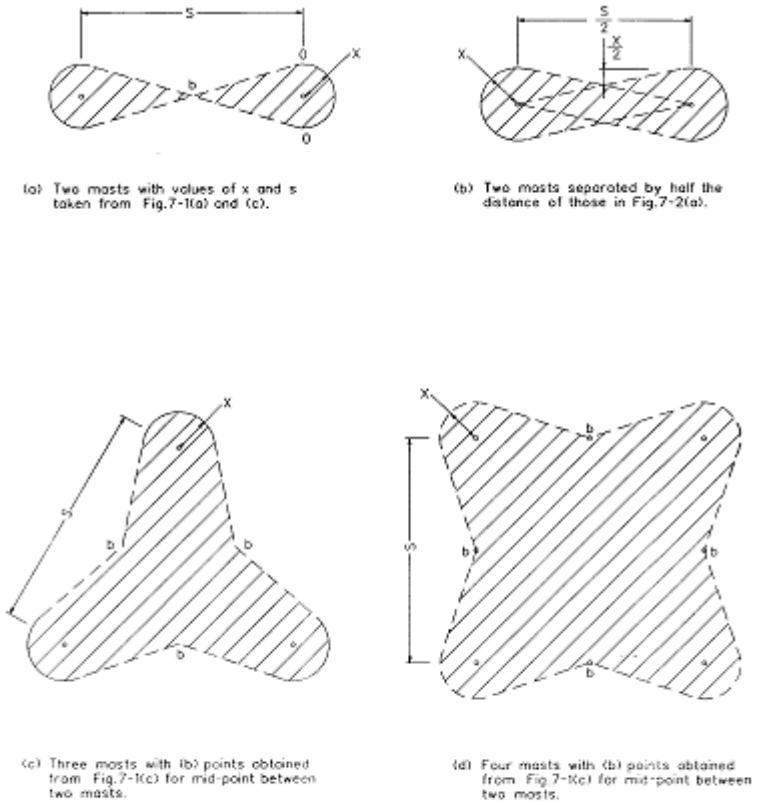

Figure 07-2: Areas Preotected by Multiple Masts for Point Exposure of

0.1 Percent

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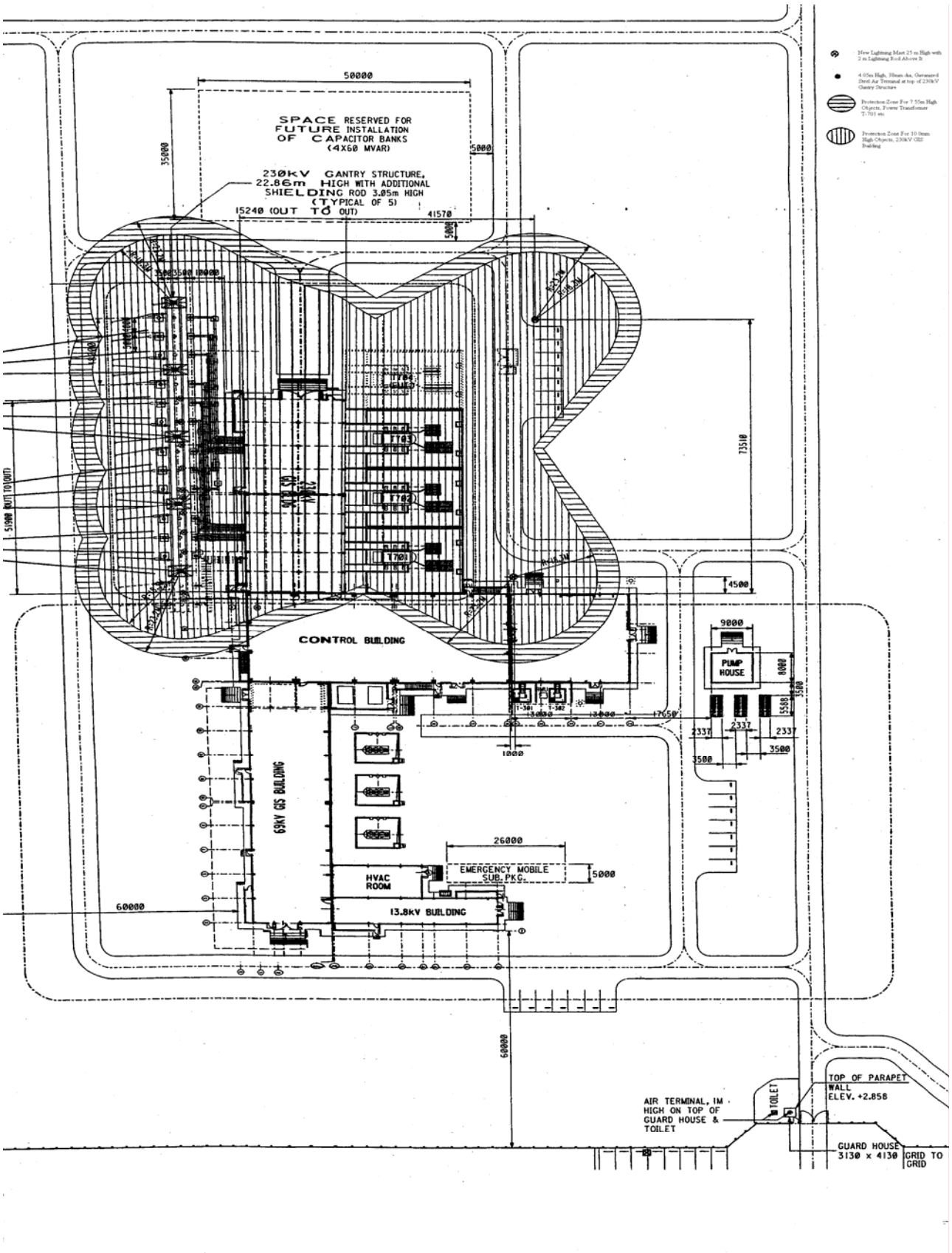

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Figure 07-3: Sample Layout for Illustration of the Use of Curves of Figure 07-1 and Configuration of Figure 07-2

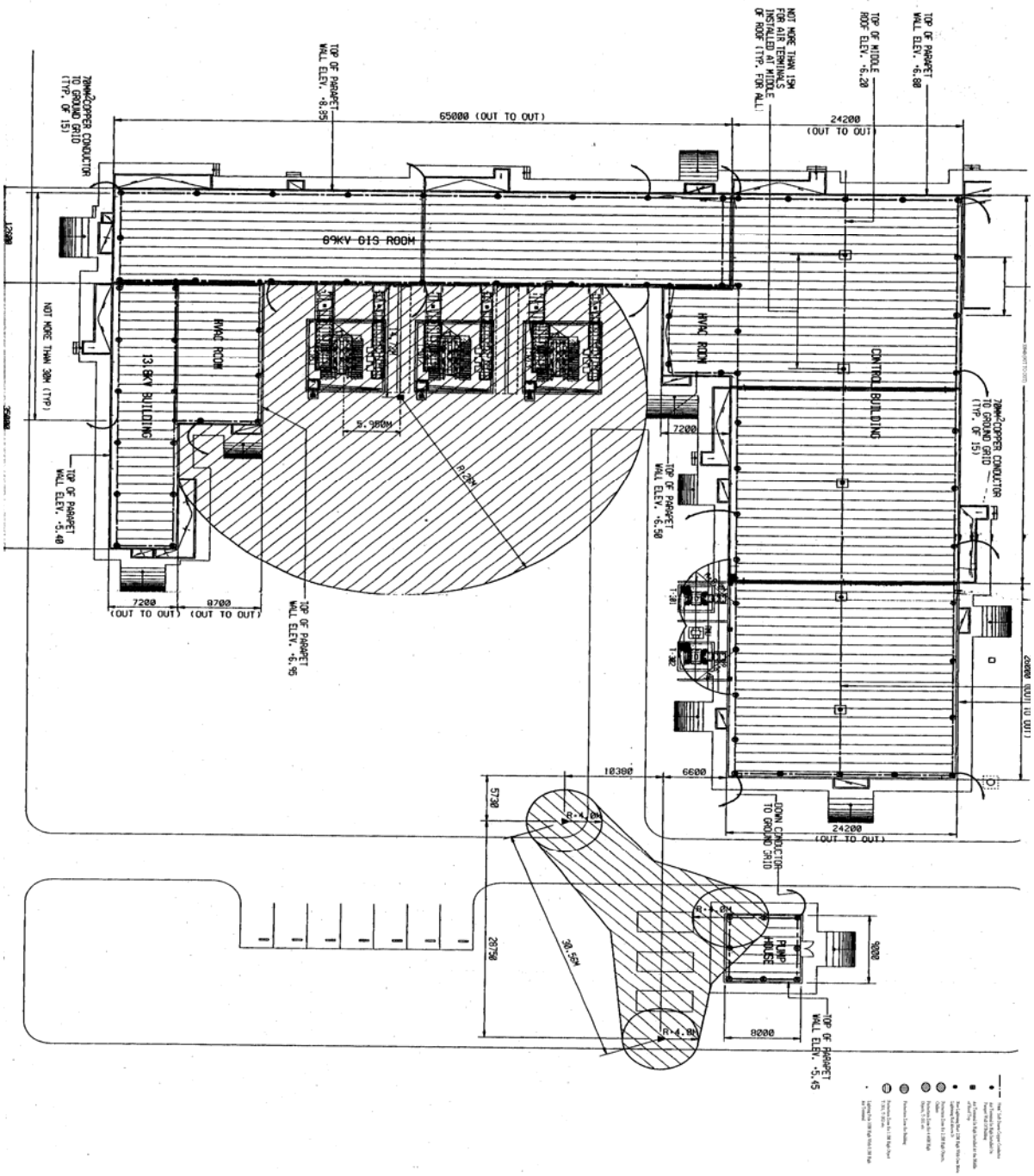

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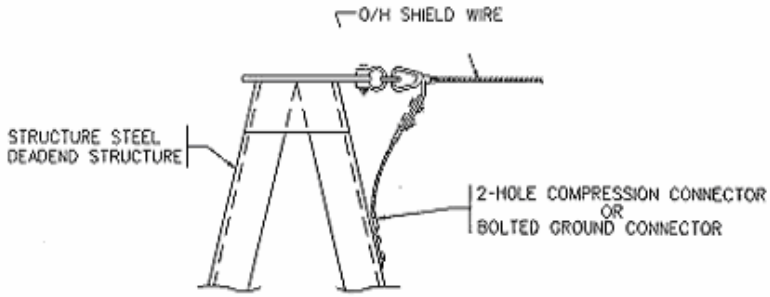
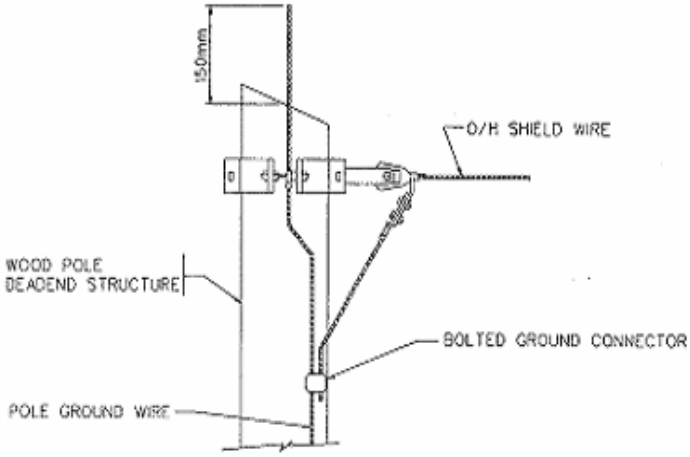
Figure 07-4: Sample Layout for Illustration of the Use of Curves of Figure 07-1 and Configuration of Figure 07-2



(a) SHIELD WIRE CONNECTIONS
STRUCTURRL STEEL DEADEND STRUCTURE



(b) SHIELD WIRE CONNECTION
WOOD POLE DEADEND STRUCTURE

Figure 07-5: Shield Wire Connection to Dead End Structure

9.0 BIBLIOGRAPHY

1. ANSI/IEEE 141: "Recommended Practice for Electric Power Distribution for Industrial Plants".

2. Khalil Denno, "High Voltage Engineering in Power Systems", CRC Press, Inc., USA, 1992.

3. NFPA 780: "The Installation of Lightning Protection System",.

4. REA Bulletin 65-1, "Design Guide for Rural Substations", 1978.

5. Westinghouse Electric Corporation, "Electrical Transmission and Distribution Reference Book", Fourth Edition, Tenth Printing, Pennsylvania, USA, 1964.

6. IEEE 998 “Guide for Direct Lightning Stroke Shielding of Substations”.